RAJALAKSHMI ENGINEERING COLLEGE

Vignesh K 250701825-2025-CSE

V2

Started on	Wednesday, 12 August 2026, 8:50 AM
State	Finished
Completed on	Friday, 21 August 2026, 8:28 AM
Time taken	8 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int find(int arr[],int low,int high,int x){
    if(low>high){
        return -1;
    }
    if(x>=arr[high]){
        return arr[high];
    }
    int mid = low+(high-low)/2;
    if(arr[mid]==x){
        return arr[mid];
    }
    if(mid>0 && arr[mid-1]<=x && x<arr[mid]){
        return arr[mid-1];
    }
    if(x<arr[mid]){
        return find(arr,low,mid-1,x);
    }
    return find(arr,mid+1,high,x);
}

int main(){
    int n,x;
    if(scanf("%d",&n)!=1)
        return 0;
    int arr[n];
    for(int i=0;i<n;i++){
        if(scanf("%d",&arr[i])!=1){
            return 0;
        }
    }
    if(scanf("%d",&x)!=1){
        return 0;
    }
    int flooor=find(arr,0,n-1,x);
    printf("%d\n",flooor);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course